Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

The concepts from class and previous assignments that will help me complete this project are the use of routes and pages within a drafter website. Dataclasses will be helpful. Textboxes will be very helpful when naming buttons or other parts in the website. I definitely expect to use a good amount of if statements throughout my code. For the loops there will definitely be a for loop used somewhere. The programming skill I want to improve through doing this project is using drafter and building functional websites.

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

The part I expect to be the easiest is making the buttons on the page. We have good examples of this being done in previous assignments and the previous project as well. The hardest part will be getting certain things on the screen to bring you to the new page when pressed. I do not have any memory of how this worked.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

I think the thing that will take the most time and effort is coding the math that will be used to add all the inputted values together. The features or components that should be implemented first are the drafter and pages along with the buttons that will bring you to each new page.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

Some things that may distract me are my phone or the work I have to do for my other classes. I will minimize my phone distracting me by putting it on do not disturb and keeping it on the other side of the room. I do not have any other questions.

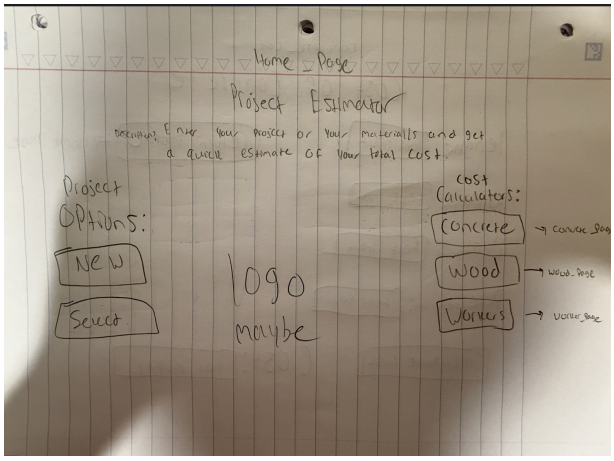
1) Site Diagram

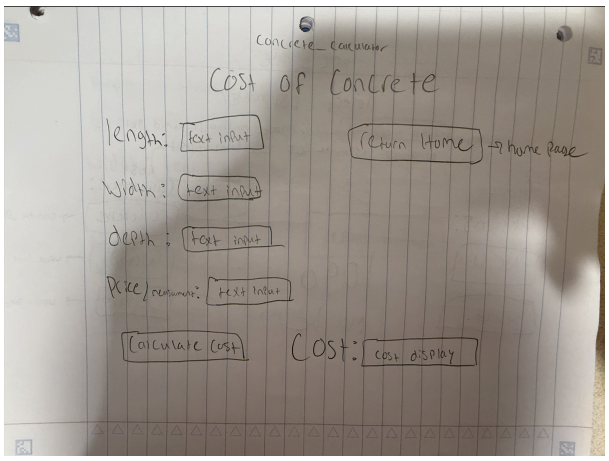
Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

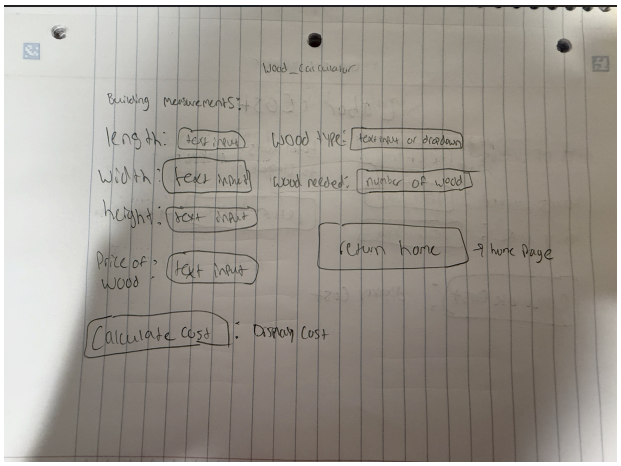
We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

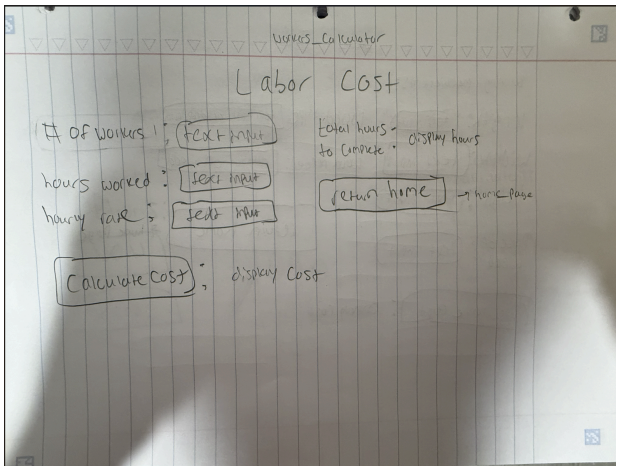
In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

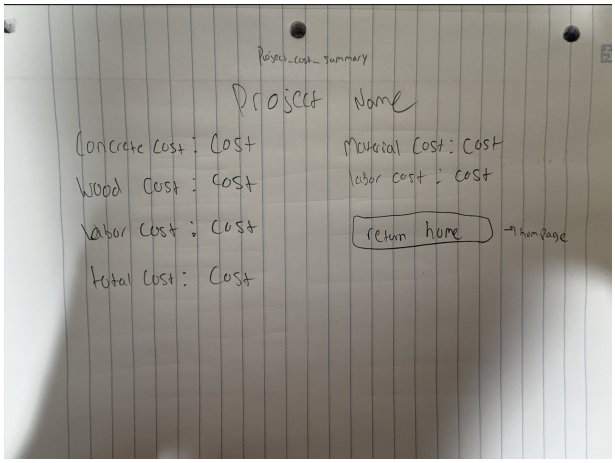
Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.


Home_Page
Project Estimator
Project Options:
New
Select
logo
maybe
Cost Calculators:
Concrete
Wood
Workers


Concrete_Calculator
Cost of Concrete
length:
Width:
depth:
Price/cubic yard:
Calculate Cost
Cost:
cost display
return Home


Wood_Calculator
Building measurements:
length:
Width:
height:
Price of wood:
Wood type:
wood needed:
Calculate cost
return home


Workers_Calculator
Labor Cost
of workers:
hours worked:
hourly rate:
Calculate cost
total hours to complete:
display hours
return home


Project_cost_summary
Project Name
Concrete cost: cost
Wood cost: cost
Labor cost: cost
total cost: cost
Material cost: cost
labor cost: cost
return home

2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	The purpose/functionality of my site is to quickly estimate the cost of small construction projects for people. They will choose what type of project they are working on and input the prices of the materials needed for that project then the total cost will be given to them.
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	current_page: str – Tracks which page the user in on, changes when page changes project_name: str – stores what project the user is working on, changes depending on what user inputs selected_material: str – stores the name of the material being calculated, changes depending on what the user selects estimates: list[Estimate] – stores a list of Estimate dataclass total_cost: float – stores combined cost of the material estimates, changes whenever a new material is added or taken away
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	Estimate dataclass material_name: str – name of material quantity: float – estimated amount needed unit: str – measurement unit unit_price: float – the price for a unit of material total_cost: float – total estimated cost for the material
Routes (minimum 7)	home_page(state): displays home page

The names of your route functions	project_details_page(state) – allows the user to select their project lumber_page(state) – displays the calculator for lumber cost plywood_page(state) – user can calculate the cost of plywood needed for the project roofing_page(state) – estimates cost of roofing materials concrete_page(state)- estimates cost of concrete needed workers_page(state) – estimates cost of the labor summary_page(state) – shows results of combined estimation cost for whole project
Pages (minimum 5) A brief description of the pages in your site	Home page: introduces you to the website and gives the user the option to choose what project or material they want to estimate the cost of Concrete calculator: user can enter the dimensions needed and calculate the cost of needed concrete Wood calculator: displays a calculator that adds the cost of plywood and lumber needed for the project Workers calculator: displays a calculator for the estimated cost of labor depending amount of workers needed and complexity of project Project cost summary: displays all of the saved estimate costs, quantities, and the combined total cost of the project.
User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these input types, see Drafter Documentation.	One text box would be for the user to input which project they are working on. I want to also look into using a SelectBox for that but I do not know how they are used. A good amount of text boxes would be used for having the user input their known cost of each material into. A third TextBox or SelectBox could be used to ask the user if they know what project they want to do or if they know the budget, workers, materials, and time they have.

If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	If statements will be used to calculate the cost of total materials as a whole and individually too. Also an if statement will be used to give the needed materials based on what project is chosen.
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in your State attribute. Description of this loop and why it is meaningful to the site	The loop would be something like for each material in the project, then it would have if statements for certain materials and inside the if statements is how much to add to the cost.

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

I am working on coding a website that is a construction cost estimator. Can you do an example of a site diagram that has the pages: Home page, concrete calculator page, wood calculator page, workers calculator page, and project cost summary page?

Page 1: Home Page (`home_page`)

- BuildCalc logo and website name.
- Short introduction explaining the purpose of the website.
- Buttons for Concrete Calculator, Wood Calculator, and Workers Calculator.
- Button to start a new construction project.
- Navigation to the other pages.

Page 2: Concrete Calculator (`concrete_page`)

- Input for concrete length.
- Input for concrete width.
- Input for concrete depth.
- Input for concrete price per cubic yard.
- Calculate button.
- Display estimated concrete volume and total cost.

- Button to save estimate and return to homepage or view project summary.

Page 3: Wood Calculator (`wood_page`)

- Input for building length, width, and height.
- SelectBox to choose lumber type (2x4, 2x6, 4x4, etc.).
- Input for lumber price.
- Calculate button.
- Display estimated amount of lumber needed and total cost.
- Button to save estimate and return to homepage or view project summary.

Page 4: Workers Calculator (`workers_page`)

- Input for number of workers.
- Input for estimated hours worked.
- Input for hourly pay per worker.
- Calculate button.
- Display total labor hours and estimated labor cost.
- Button to save estimate and return to homepage or view project summary.

Page 5: Project Cost Summary (`summary_page`)

- Display the project name.
- Display concrete material estimate and cost.
- Display wood material estimate and cost.
- Display worker labor estimate and cost.
- Calculate and display the total estimated project cost.
- Display a chart comparing material and labor expenses.
- Button to return to homepage or start a new project.

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.